

UbuntuNet Alliance Women Hackathon 2026

Phase I - Week 1 Submission

SOLARTRACE

Chapter 1: Introduction

Team South Sudan



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1 BACKGROUND TO THE STUDY

1.1 The Energy Access Crisis in Sub-Saharan Africa

Sub-Saharan Africa faces the most severe energy access deficit globally. According to the International Energy Agency [1], only **48%** of the population has access to electricity, with rural areas experiencing significantly lower rates. South Sudan, the project's target country, has one of the lowest electricity access rates globally at approximately **7%**.

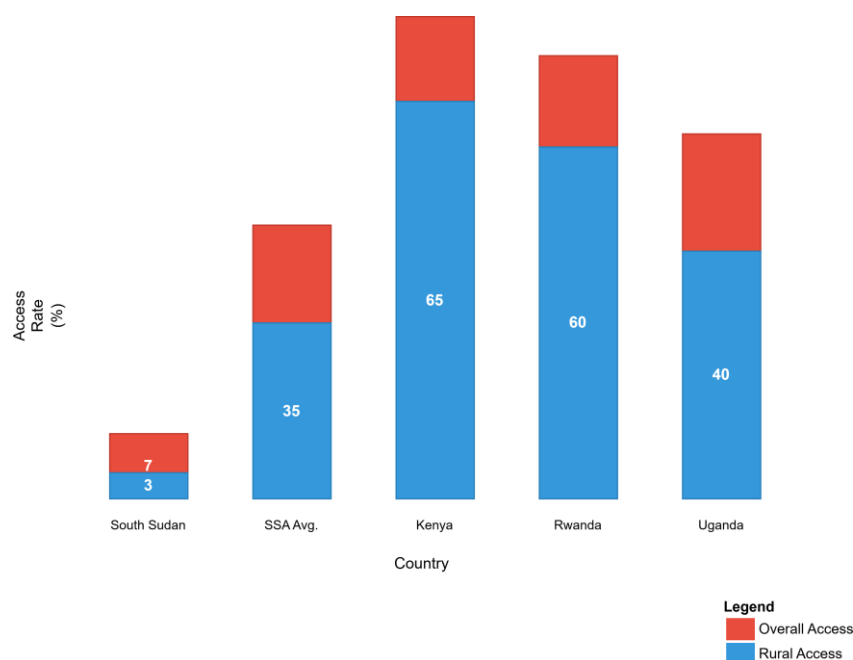


Figure 1: Electricity Access Rates in Select Countries

1.2 Biomass Dependence and Its Consequences

Across Sub-Saharan Africa, over **80% of households** depend on biomass—firewood and charcoal—for cooking meals, boiling water, and sterilizing tools. This dependence has multiple cascading consequences.

Table 1: Biomass Dependence in South Sudan - Key Statistics

Indicator	Value
Households dependent on biomass	>80%
Average monthly fuel expenditure	USD 30-50
Daily time spent collecting firewood	4-6 hours
Forest cover loss since 2000	~20%
Annual CO ₂ per household	~1.0 ton

1.2.1 Environmental Impact

The *State of the World's Forests 2024* [2] reports that over **4.5 million hectares** of forest are lost annually in Africa, with South Sudan losing approximately **20%** of its forest cover since 2000. The IPCC [3] notes that each household using biomass contributes approximately **1.0-ton CO₂ annually**, accelerating climate change and biodiversity loss.

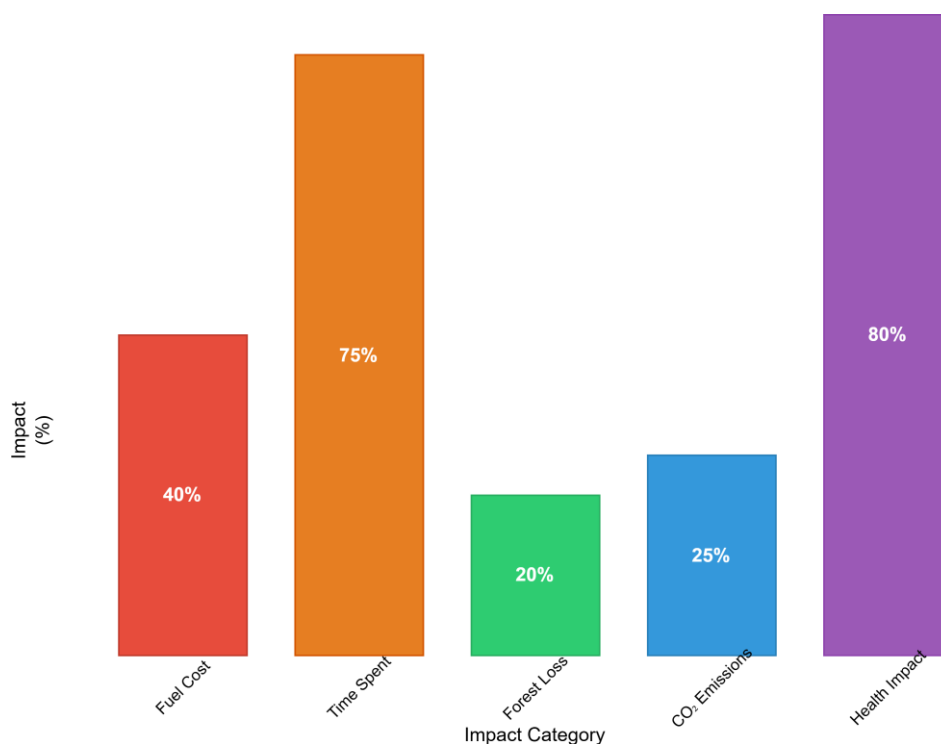


Figure 2: Consequences of Biomass Dependence

1.2.2 Economic Burden

Ndung'u and Mutua [4] document that typical families spend **USD 30-50 monthly** on charcoal or firewood, representing a significant portion of household income in rural areas. Small rural clinics often spend over **USD 100 monthly** on kerosene for sterilization. Women and children spend **4-6 hours daily** collecting firewood, reducing time for education and income-generating activities.

1.2.3 Health Impact

The World Health Organization [5] reports that indoor air pollution causes **3.2 million premature deaths** annually, with women and children disproportionately affected. Respiratory diseases, eye infections, and burns are common, with children under five being particularly vulnerable.

1.3 South Sudan Context

South Sudan presents a unique set of challenges and opportunities for solar cooking:

- **Abundant solar resource:** The country receives an average of 5-7 kWh/m²/day
- **Post-conflict context:** Over **2 million internally displaced persons** and **2.2 million refugees**
- **Fragile health system:** Many rural clinics lack reliable electricity for sterilization
- **Gender inequality:** Women bear the primary burden of fuel collection
- **Emerging ICT infrastructure:** Growing mobile phone penetration creates opportunities for IoT

Table 2: South Sudan - Key Contextual Statistics

Indicator	Value
Electricity Access Rate	7%
Daily Solar Radiation	5-7 kWh/m ² /day
Household Fuel Expenditure	USD 30-50/month
Forest Cover Loss (2000-2020)	~20%
Internally Displaced Persons	>2 million
Mobile Phone Penetration	~40%

1.4 Solar Cooking Technologies

1.4.1 The Problem with Manual Solar Cookers

While solar cookers offer a clean alternative to biomass, their effectiveness is limited by:

- **Need for constant manual repositioning:** Users must adjust the cooker every **30-60 minutes** to track the sun [6]
- **Temperature fluctuations:** Without consistent adjustment, cooking temperatures decline by **20-30°C** [7]
- **Poor user adoption:** The inconvenience of manual tracking leads to abandonment
- **Limited functionality:** Many manual cookers cannot achieve or maintain temperatures required for sterilization (120°C)

1.4.2 Advances in Solar Cooking Technology

Recent research has focused on addressing these limitations:

- **Automatic tracking systems:** Using motors and sensors to follow the sun [8, 6]
- **IoT integration:** Remote monitoring and control via mobile applications [9, 10]
- **Thermal energy storage:** Extending cooking beyond daylight hours
- **Hybrid designs:** Combining solar with alternative heat sources [7]

Mahmood et al. [7] demonstrated that tracked solar cookers achieve significantly higher temperatures, with their hybrid solar cooker achieving 95°C with tracking compared to 86°C without tracking.



Figure 3: Hybrid Solar Cooker Performance with Tracking [7]

2 STATEMENT OF THE PROBLEM

2.1 Core Problem

Rural and peri-urban communities in South Sudan lack access to clean, reliable, and affordable cooking energy, leading to deforestation, health crises, and economic hardship, while existing solar cooking solutions remain underutilized due to the need for constant manual tracking.

2.2 Problem Breakdown

1. Energy Access Crisis

- 93% of the population lacks electricity access [1]
- 80%+ of households depend on biomass (firewood/charcoal) [2]
- Fuel collection consumes 4-6 hours daily, primarily by women and children [5]

2. Environmental Degradation

- South Sudan has lost 20% of forest cover since 2000 [2]
- Deforestation driven by charcoal production and firewood collection
- Each household contributes 1.0-ton CO₂ annually [3]

3. Health Crisis

- Indoor air pollution from biomass causes 3.2 million premature deaths globally [5]
- Women and children are disproportionately affected in South Sudan
- Rural clinics lack fuel for sterilization of surgical instruments

4. Economic Burden

- Households spend USD 30-50 monthly on cooking fuels [4]
- Rural clinics spend USD 100+ monthly on kerosene
- Fuel collection reduces time for education and income generation

5. Technology Adoption Barriers

- Manual solar cookers require repositioning every 30-60 minutes
- Without consistent adjustment, temperatures decline by 20-30°C [7]
- Poor user adoption due to inconvenience and limited functionality

2.3 The Research Gap

While automatic tracking systems have been developed in controlled settings, they are not widely available or affordable in South Sudan. IoT integration for remote monitoring and control of solar cookers is limited in the region. There is no documented RTC-based tracking system designed specifically for the South Sudan context, and rural clinics lack access to clean, fuel-free sterilization technology.

Table 3: Comparison of Existing Solar Cooker Systems vs. SolarTrace

Feature	Existing Systems	SolarTrace
Sun Tracking	Manual (30-60 min intervals)	Automatic (RTC-based, 10 min)
IoT Integration	Limited or none	Full Blynk integration
Mobile Application	Rare	Comprehensive mode selection
Data Logging	Not available	SD card + cloud logging
Sterilization Mode	Not supported	120°C for 20 minutes
Safety Mechanism	None	130°C automatic shutdown
Target Region	Global focus	Designed for South Sudan

2.4 Consequences of Inaction

If this problem is not addressed:

- Deforestation will continue, accelerating climate change
- Households will remain trapped in poverty due to fuel expenditure
- Women and children will continue to suffer health impacts and lost opportunities
- Rural clinics will struggle to provide safe surgical care
- South Sudan will miss its climate mitigation and adaptation goals

3 NOVELTY AND INNOVATION

3.1 What Makes SolarTrace Novel

The innovation of SolarTrace lies in the integration of multiple technologies into a cohesive, low-cost system designed specifically for the South Sudan context:

1. **First RTC-Based Tracking System for South Sudan:** No documented system exists that uses a Real-Time Clock module for automatic sun tracking specifically calibrated for South Sudan's latitude (4.85°N)
2. **Integrated IoT and Mobile Monitoring:** Unlike manual solar cookers, SolarTrace provides real-time temperature monitoring and push notifications via the Blynk platform
3. **Dual-Purpose Design:** The system supports both cooking and sterilization applications, addressing both household and clinic needs
4. **Offline Data Logging:** SD card storage ensures data collection even in areas without internet connectivity
5. **Low-Cost Implementation:** Bill of Materials under USD 300 makes the system accessible for local manufacturing

3.2 Comparison with Existing Work

Table 4: SolarTrace Innovation Comparison

Aspect	Abid (2017)	Mahmood et al. (2025)	SolarTrace
Tracking Type	Dual-axis	Hybrid (box + tube)	Single-axis RTC-based
IoT Platform	Not specified	Not specified	Blynk
Mobile App	No	No	Yes
Sterilization Mode	No	No	Yes (120°C/20 min)
Safety Mechanism	No	No	Yes (130°C shut-down)
Target Region	Generic	Iraq	South Sudan
Cost	Not specified	Not specified	< USD 300

3.3 Expected Improvements

Based on literature estimates [11, 12], SolarTrace is expected to achieve:

- **40-60% reduction** in household cooking fuel costs (literature-based estimate)

- **30-40% improvement** in cooking temperature consistency compared to manual tracking
- **USD 100+ /month savings** for rural clinics through fuel-free sterilization
- **1.0-ton CO₂ savings** per household annually (based on IPCC [3] estimates)
- **4-6 hours/day** of time freed for women and children

4 PROTOTYPE OBJECTIVES

4.1 General Objective

To develop and pilot a smart, IoT-connected parabolic solar cooker with time-based automatic sun tracking that reduces household cooking energy costs by at least 40% compared to charcoal.

4.2 Specific Objectives

1. **Implement RTC-triggered servo movement** every 10 minutes based on pre-calculated solar angles for the local latitude (4.85°N), achieving tracking accuracy within $\pm 5^\circ$
2. **Develop a user-friendly mobile application** using Blynk that allows selection of cooking modes and provides real-time temperature monitoring
3. **Deliver push notifications** when target temperatures are achieved or when unexpected temperature drops occur
4. **Record each cooking session**, including duration, peak temperature, and estimated CO₂ savings, with offline SD card storage capability
5. **Automatically tilt the reflector away from the sun** when temperatures exceed 130°C to prevent equipment damage and ensure user safety

Table 5: Prototype Objectives and Corresponding Technologies

Objective	Technology	Reference
RTC-based sun tracking	DS3231 RTC, MG995 Servo	[8, 13]
Mobile application	Blynk IoT Platform	[9]
Push notifications	Blynk Notifications	[10]
Data logging	MicroSD card module	[9]
Temperature monitoring	MAX6675 K-type thermocouple	[9]

4.3 System Architecture

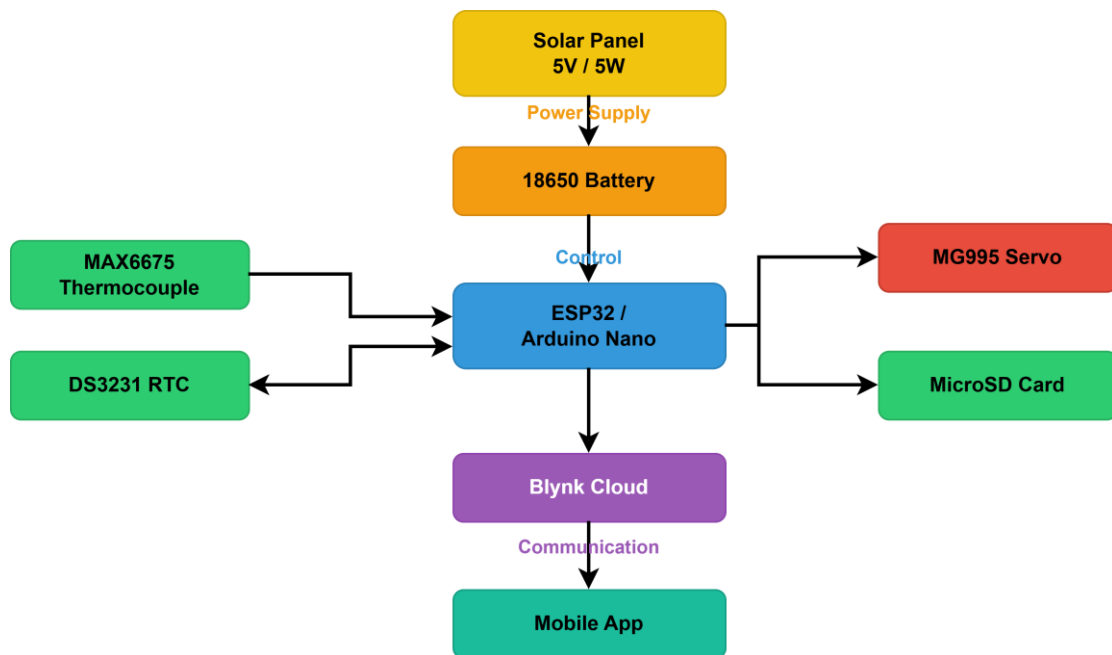


Figure 4: SolarTrace System Architecture

5 PROTOTYPE RESEARCH QUESTIONS

1. **Tracking Accuracy:** How accurate is the time-based tracking algorithm for Juba, South Sudan (4.85°N) when compared to optimal solar positioning?
2. **User Interface Design:** How can a Blynk-based mobile interface be designed to provide intuitive control and monitoring for users with basic digital literacy?
3. **Notification Effectiveness:** Can IoT-based push notifications improve user experience and reduce the risk of undercooked food or failed sterilization?
4. **Data Logging Capability:** What is the effectiveness of an SD card-based logging system for capturing and analyzing cooking data in off-grid environments?
5. **Safety Mechanism Reliability:** Does the safety mechanism trigger reliably at 130°C and prevent overheating in all operational conditions?

6 SIGNIFICANCE OF THE STUDY

6.1 Who Benefits?

Table 6: Stakeholder Impact Matrix

Stakeholder	Primary Benefit	Scale
Rural Households	Reduced fuel expenditure (40-60%), reduced smoke exposure	Direct
Women & Children	Reduced time collecting fuel (4-6 hours/day), improved health	Direct
Rural Clinics	Fuel-free sterilization, cost savings (USD 100+/month)	Direct
Community	Reduced deforestation, improved environmental health	Indirect
Government	Progress toward climate goals (NDCs)	Indirect
Environment	Reduced CO ₂ emissions (1.0 ton/household/year)	Global

6.2 Technical Significance

- **First RTC-based automatic tracking system designed for South Sudan**
- **Integration of IoT (Blynk) for remote monitoring and control**
- **Open-source documentation for replication and adaptation**
- **Low-cost design (BOM ; USD 300) suitable for local manufacturing**

6.3 Scientific Significance

- Contributes to the growing body of research on IoT in renewable energy [14, 15]
- Provides empirical data on solar cooking in a post-conflict context
- Advances understanding of technology adoption in low-resource settings
- Documents the feasibility of solar sterilization in rural clinics [16]

6.4 Policy Significance

- Supports South Sudan's Nationally Determined Contributions (NDCs) under the Paris Agreement
- Aligns with SDG 7 (Affordable and Clean Energy), SDG 13 (Climate Action), and SDG

5 (Gender Equality)

- Provides evidence for renewable energy policy development
- Demonstrates the potential of ICT in climate adaptation

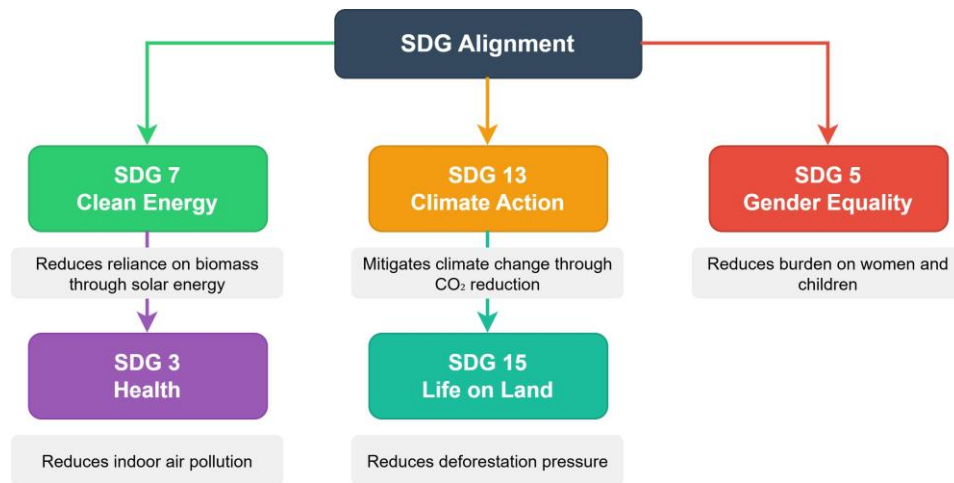


Figure 5: SolarTrace Alignment with Sustainable Development Goals

7 DELIMITATIONS OF THE STUDY

7.1 Geographic Scope

- **Primary focus:** Juba, South Sudan (4.85°N latitude)
- **Reason:** Capital city with existing ICT infrastructure and research partners
- **Exclusion:** Other regions of South Sudan (pending validation)

7.2 Technical Scope

- **Prototype only:** Designed for demonstration and pilot testing
- **Hardware:** Parabolic reflector (0.9m), Arduino Nano/ESP32, DS3231 RTC, MAX6675 thermocouple, MG995 servo
- **Software:** C++ (Arduino IDE), Blynk IoT platform
- **Exclusion:** Large-scale production or commercialization (beyond hackathon scope)

Table 7: Technical Specifications and Delimitations

Component	Specification	Limitation
Parabolic Reflector	0.9m diameter	Fixed size, not scalable
Microcontroller	Arduino Nano/ESP32	Limited processing power
RTC Module	DS3231	Requires battery backup
Temperature Sensor	MAX6675 K-type	0-500°C range
Actuator	MG995 Servo	Torque limited to 15kg/cm

7.3 Functional Scope

- **Cooking modes:** Four modes (cooking, sterilization, boiling, warming)
- **Temperature range:** 70-130°C (with safety cut-off at 130°C)
- **Application:** Cooking staple foods, boiling water, sterilization
- **Exclusion:** Space heating, water heating for bathing, electricity generation

7.4 Temporal Scope

- **Prototype development:** July-August 2026 (Hackathon Phase I)
- **Testing:** September 2026 (Phase II preparation)
- **Evaluation:** October 2026 (Phase II - Malawi)
- **Exclusion:** Long-term impact assessment (beyond hackathon timeline)

7.5 User Scope

- **Primary users:** Rural and peri-urban households, small rural clinics
- **User profile:** Basic digital literacy assumed
- **Exclusion:** Large-scale institutional or commercial cooking

7.6 Limitations (Acknowledged Constraints)

- **Weather dependence:** System requires adequate sunlight (cloudy days affect performance)
- **Power supply:** Requires 5V solar panel with battery backup
- **Time constraints:** Hackathon duration limits extensive user testing
- **Resource constraints:** No funding for large-scale field trials
- **Technical expertise:** Team members have varying programming proficiency

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TEAM SOUTH SUDAN - SOLARTRACE

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